

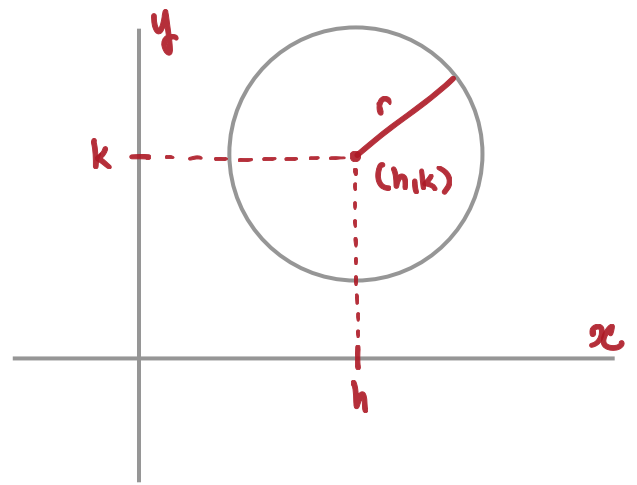
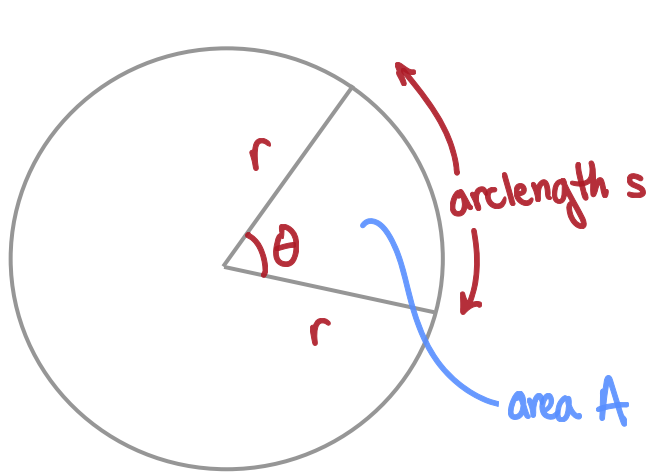
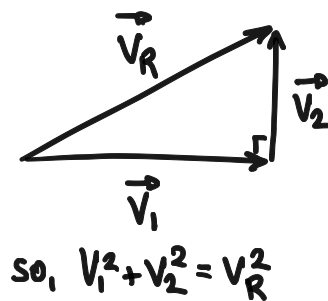
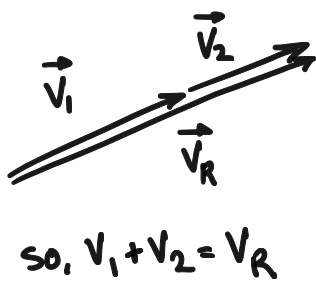
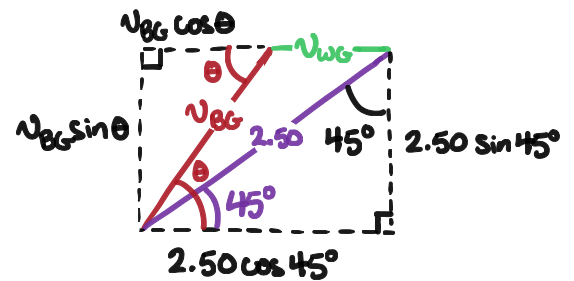
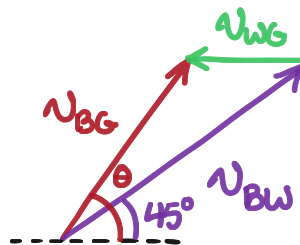
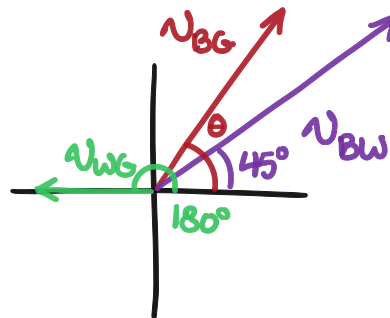
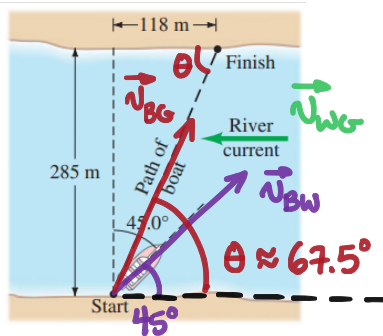
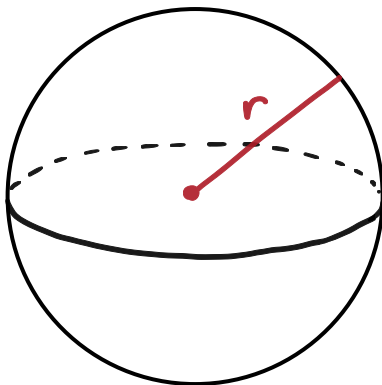
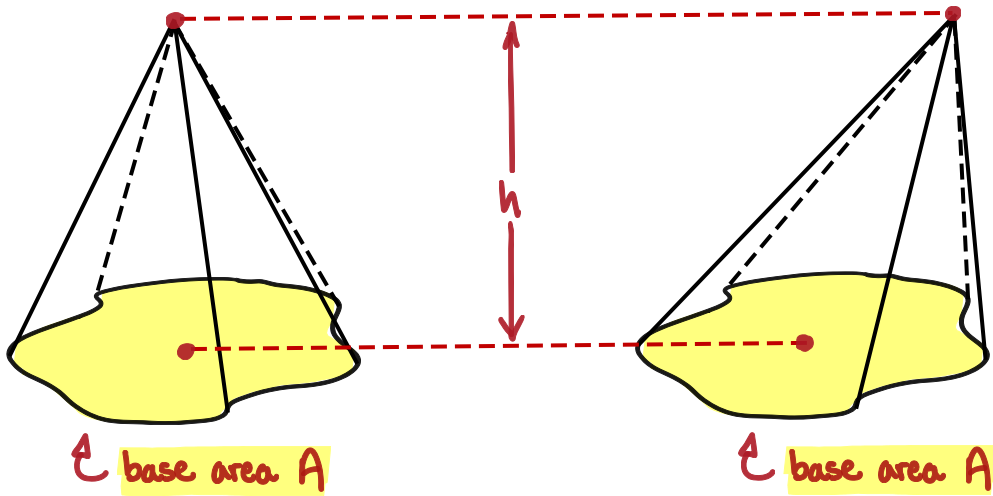
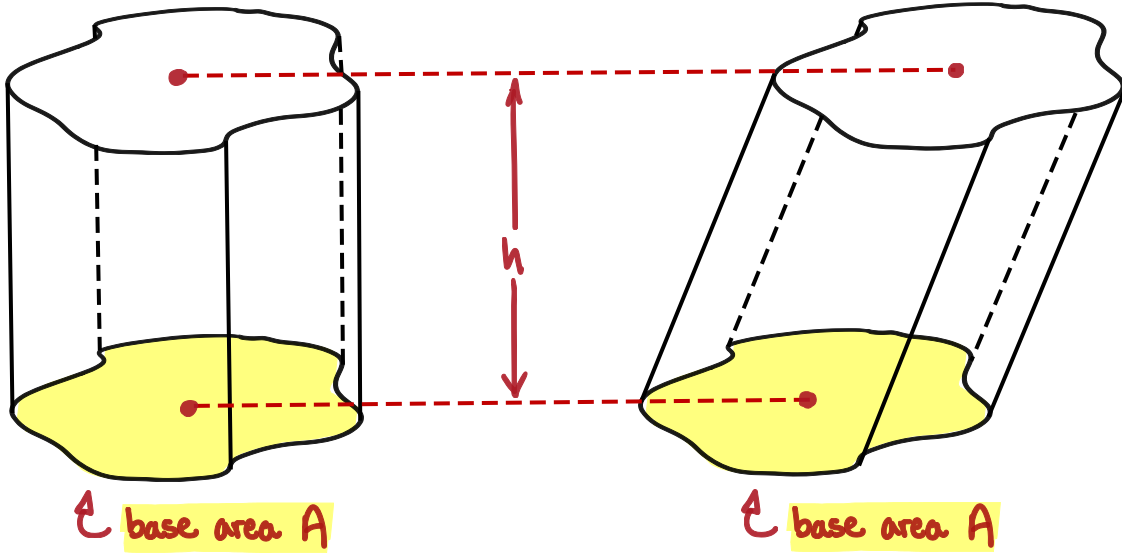
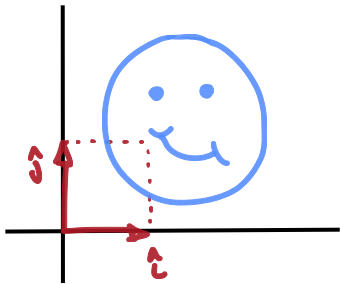
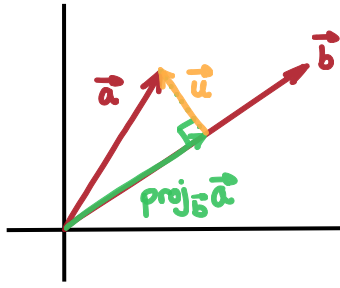
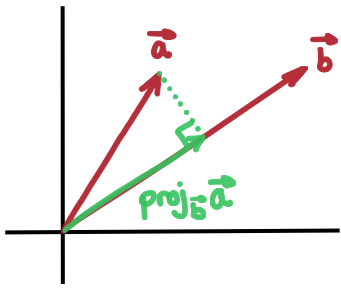


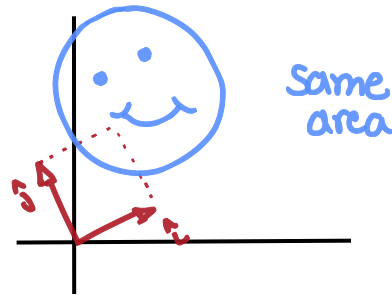
FIGURE 3-45
Problem 48.





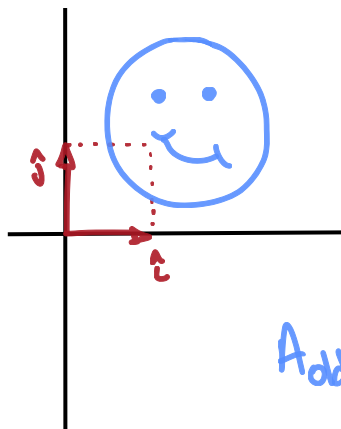


M
rotation

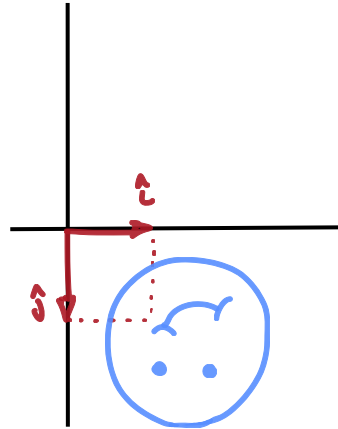


$$A_{old} \cdot 1 = A_{new}$$

$\uparrow \det M = 1$



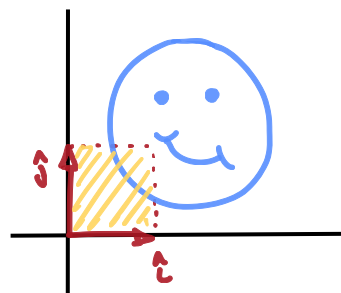
M
x-axis
reflection



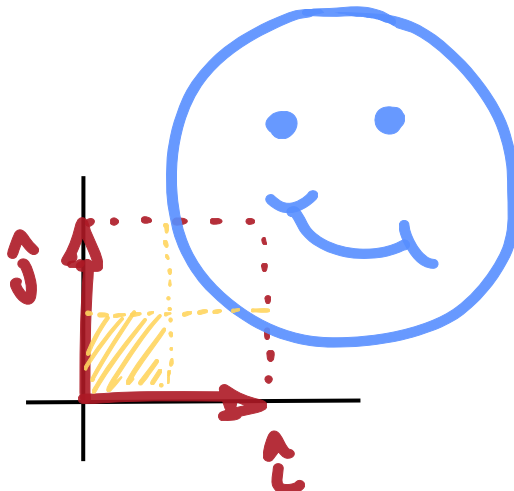
$$A_{old} \cdot 1 = A_{new}$$

$\uparrow \det M = -1$

(minus sign indicates flipped image)



M
 $\times 2$ in x
and y directions



$$A_{old} \cdot 4 = A_{new}$$

$\uparrow \det M = 4$

